The Exact Log-Concavity Threshold for Partitions into Squares

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Abstract

Let $s(n)$ count unordered partitions of n into positive squares, with unlimited repetitions. We prove that $s(n)^2 > s(n-1)s(n+1)$ for every $n \geq 1042$, while the inequality fails at 1041. Thus the exact onset of permanent nonnegative log-concavity is 1042. We also determine all 421 negative indices and all six equality indices. The proof combines an explicit analytic estimate valid beyond 1,200,000 with a complete exact-arithmetic verification of the adjoining finite interval. The analytic argument uses an exact transformation of the generating function, a tenth-order approximation on a steepest contour with proved remainder bounds, and separately certified minor-arc estimates. Its coefficient errors are bounded before taking second differences; no uncontrolled remainder is differentiated. All finite certificates and verifiers are included in the accompanying reproducibility package.

1 The result and its scope

The square-partition function is defined by the formal generating series

$$\mathcal{F}(q) = \sum_{n\geq 0} s(n)q^n = \prod_{j\geq 1}(1-q^{j^2})^{-1}, \qquad s(0) = 1. \tag{1}$$

Thus parts are positive perfect squares, each may occur any number of times, and order is ignored. For example $s(5) = 2$, corresponding to $4+1$ and five copies of 1. This function differs from the number of representations by a fixed number of squares and from partitions into distinct squares.

For $n \geq 1$ put

$$D(n) = s(n)^2 - s(n-1)s(n+1), \qquad T = \min\{t \in \mathbb{Z}_{\geq 1} : D(n) \geq 0 \text{ for every integer } n \geq t\}. \tag{2}$$

For integers $a \leq b$ of the same residue modulo four, write $[a,b]_4 = \{a, a+4, \ldots, b\}$.

Theorem 1.1. *The exact threshold in (2) is $T = 1042$. More precisely, $D(n) > 0$ for every integer $n \geq 1042$. The set of all positive integers with $D(n) < 0$ is exactly the 421-element set in Table 1. The equality set is*

$$\{n \geq 1 : D(n) = 0\} = \{1, 2, 5, 6, 10, 14\}.$$

At every other positive integer, $D(n) > 0$.

The sharp boundary is visible in the exact values

$$D(1041) = -5{,}695{,}922{,}185{,}615 < 0, \tag{3}$$

$$D(1042) = 154{,}125{,}878{,}482{,}603 > 0. \tag{4}$$

These computations alone do not prove permanence. The infinite range is covered by the analytic argument below, and every remaining integer is covered by the exact finite computation in Section 7.

Residue	Negative indices	Count
0 (mod 4)	$[24, 28]_4$, $[44, 48]_4$, $[56, 64]_4$, $[76, 644]_4$, $[652, 656]_4$	152
1 (mod 4)	$\{69, 105, 177, 201\}$, $[213, 217]_4$, $[229, 237]_4$, $[245, 257]_4$, $[265, 1041]_4$	208
2 (mod 4)	None	0
3 (mod 4)	$[3, 219]_4$, $[227, 235]_4$, $[247, 251]_4$, $\{267\}$	61

Table 1: The complete negative set, grouped into disjoint arithmetic progressions and singleton sets. The grouping is a lossless compression of individually checked indices.

1.1 Prior work and the indexing convention

Eventual log-concavity of partitions into powers is known. O'Sullivan [3] proves detailed arbitrary-order asymptotic expansions and strict eventual log-concavity; Benfield, Paul and Roy [4] also prove eventual Turán inequalities. For squares the normalized second difference has leading scale $n^{-5/3}$. An expansion with an unspecified error constant or an unspecified validity cutoff cannot by itself connect this fact to a particular finite computation.

Roy's journal article [1], Section 3.1, Conjecture 3.3 and Table 3, gives $N_2 = 1041$ under the convention $n > N_2$. The convention here is $n \geq T$, so the square case is exactly $T = 1042$. The journal version, published on 29 September 2026, and arXiv version 4 [2] both list Roy alone; versions 1–3 of that preprint list Brennan Benfield and Arindam Roy. OEIS A346160 [6] provides related numerical evidence, but its finite computation and its occasionally inconsistent example wording do not establish the infinite tail or determine our indexing.

A literature search on 23 October 2026 examined the cited journal and preprint versions, later publications, equivalent formulations involving Turán inequalities. No prior proof of the sharp square-case threshold was located. This is a statement about the documented search, rather than a guarantee concerning unindexed work. The contribution here is the explicit sharp threshold for squares and the complete exceptional set. It does not settle the higher-power cases of Roy's conjecture.

The proof uses Zagier's generalized eta transformation [5], whose needed specialization is derived in Section 2. This is an exact identity for the square-partition product, not an assumption of the modular transformation law for ordinary partitions. Separate estimates for the other arcs remain essential.

1.2 Constants, ranges, and the proof architecture

Write $\mathcal{F}(e^{-z})$ for the generating function in the half-plane $\operatorname{Re} z > 0$, and set

$$A = \Gamma(3/2)\zeta(3/2) = \frac{\sqrt{\pi}}{2}\zeta(3/2), \qquad a = (A/2)^{2/3}, \qquad N_0 = 1{,}200{,}000. \tag{5}$$

For real $n > 0$ define

$$t = (A/(2n))^{2/3}, \qquad x = t^{-1/2}, \qquad X = nt = \frac{A}{2}x = an^{1/3}. \tag{6}$$

The rational constant bounds used throughout are

$$\frac{231}{100} < A < \frac{58}{25}, \qquad \frac{157}{50} < \pi < \frac{22}{7}, \qquad a^6 > \frac{17}{10}. \tag{7}$$

Their interval-arithmetic proofs and exact verification are described in the certificate appendix. In particular, if $n \ge N_0$, then

$$t \le 10^{-4}, \qquad x \ge 100, \qquad X > 115. \tag{8}$$

Indeed $A/2 < 1.16$ and $n \ge 1.2 \cdot 10^6$ imply $(A/(2n))^2 < 10^{-12}$, and $X = Ax/2 > 115.5$.

The principal approximation is

$$M(n) = \frac{ae^{3an^{1/3}}}{2\sqrt{3}\,\pi^{3/2}n^{7/6}}. \tag{9}$$

We construct a rational polynomial in X^{-1}, denoted $P(X)$, with ten correction terms. The exact transformation controls the replacement of $\mathcal{F}(e^{-z})$ by its principal model on $|\operatorname{Im} z| \le t$. A steepest-contour argument controls the model integral, and elementary product estimates control the remaining arcs. Together these give an error smaller than $X^{-5}/10$ after normalization by $M(n)$. The explicit function $Q(n) = M(n)P(an^{1/3})$ has a negative second logarithmic derivative of sufficient size to dominate this coefficient error. This proves $D(n) > 0$ for every $n \ge N_0 + 1$.

2 An exact transformation with an explicit remainder

An exact transformation, rather than an asymptotic expansion with an unspecified remainder, is useful on the principal arc. The identity below is the square case of Zagier's transformation [5]. We give its Mellin proof to fix the branches, signs, and constants.

Lemma 2.1. *For* $\operatorname{Re} z > 0$*, use the principal logarithm and square root, and put*

$$A = \frac{\sqrt{\pi}}{2}\zeta(3/2), \qquad K_\pm = 2\pi^{3/2}(1 \pm i).$$

Then

$$\log \mathcal{F}(e^{-z}) = \frac{A}{\sqrt{z}} + \frac{1}{2}\log z - \log(2\pi) + R(z), \tag{10}$$

where

$$R(z) = \sum_{m,l\ge 1} \frac{e^{-K_+ l\sqrt{m/z}} + e^{-K_- l\sqrt{m/z}}}{l} \tag{11}$$

converges absolutely. The logarithm on the left is defined by the absolutely convergent logarithmic series of the Euler product.

Proof. For $c > 1/2$, Mellin inversion and absolute convergence give

$$\log \mathcal{F}(e^{-z}) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \Gamma(s)\zeta(1+s)\zeta(2s)z^{-s}\,ds. \tag{12}$$

Indeed, expand the logarithm as $\sum_{j,l\ge 1} e^{-lj^2 z}/l$ and apply Mellin inversion to each exponential. The resulting Dirichlet sums converge absolutely on this line.

Shift the line of integration to $\operatorname{Re} s = -3/2$. The crossed poles are $s = 1/2$ and $s = 0$. The apparent pole at $s = -1$ is removable because of the trivial zero of $\zeta(2s)$. The first residue is $Az^{-1/2}$. At zero,

$$\Gamma(s)\zeta(1+s) = s^{-2} + O(1), \qquad \zeta(0) = -\tfrac{1}{2}, \qquad \zeta'(0) = -\tfrac{1}{2}\log(2\pi),$$

so the residue is $\frac{1}{2}\log z - \log(2\pi)$. For completeness, the contour shift is valid for every fixed $|\arg z| < \pi/2$: on the fixed vertical strip the zeta factors have at most polynomial growth, and Stirling's formula gives exponential decay $e^{-\pi|\operatorname{Im} s|/2}$ for the gamma factor. The factor z^{-s} leaves exponential decay since $|\arg z| < \pi/2$. Thus the horizontal connections tend to zero.

In the remaining integral set $w = -2s$, taking account of the reversal of orientation and the factor $1/2$. The zeta functional equation and the gamma reflection formula give

$$\begin{aligned}\zeta(1-w/2) &= 2^{1-w/2}\pi^{-w/2}\cos(\pi w/4)\Gamma(w/2)\zeta(w/2),\\ \zeta(-w) &= -2^{-w}\pi^{-w-1}\sin(\pi w/2)\Gamma(1+w)\zeta(1+w),\\ \Gamma(-w/2)\Gamma(w/2) &= -\frac{2\pi}{w\sin(\pi w/2)}.\end{aligned}$$

Consequently, as a meromorphic identity,

$$\begin{aligned}&\tfrac{1}{2}\Gamma(-w/2)\zeta(1-w/2)\zeta(-w)\\ &\quad = 2(8\pi^3)^{-w/2}\cos(\pi w/4)\Gamma(w)\zeta(1+w)\zeta(w/2).\end{aligned}$$

Since $K_+^{-w} + K_-^{-w} = 2(8\pi^3)^{-w/2}\cos(\pi w/4)$, the remaining integral equals

$$\frac{1}{2\pi i}\int_{3-i\infty}^{3+i\infty}\Gamma(w)\zeta(1+w)\zeta(w/2)(K_+^{-w}+K_-^{-w})z^{w/2}\,dw.$$

On this line expand the two zeta factors as $\sum_{l,m\ge1} l^{-1-w}m^{-w/2}$. The absolute sum is $\sum l^{-4}m^{-3/2} < \infty$. Moreover,

$$\arg(K_\pm/\sqrt{z}) = \pm\pi/4 - \arg(z)/2 \in (-\pi/2, \pi/2).$$

Mellin inversion applies once more, separately for each sign and summand. The exponential decay of the integrand and the preceding convergent sum justify the interchange, giving exactly (11). □

Lemma 2.2. *For* $\operatorname{Re} z > 0$*, set* $\theta = \arg z$ *and*

$$b(z) = 2\pi^{3/2}|z|^{-1/2}\big(\cos(\theta/2) - |\sin(\theta/2)|\big) > 0.$$

Then

$$|R(z)| \le E(b(z)), \qquad E(b) = \frac{2e^{-b}}{1-e^{-b}}\left(1+\frac{2}{b}+\frac{2}{b^2}\right) \quad (b>0). \tag{13}$$

In particular, $|e^{R(z)} - 1| \le e^{E(b(z))} - 1$.

Proof. The number $b(z)$ is $\min_\pm \operatorname{Re}(K_\pm/\sqrt{z})$. Taking absolute values in (11), using $1/l \le 1$, and summing first in l, gives

$$|R(z)| \le \frac{2}{1-e^{-b}}\sum_{m\ge1} e^{-b\sqrt{m}}.$$

The summand decreases with m, so

$$\sum_{m\ge1} e^{-b\sqrt{m}} \le e^{-b} + \int_1^\infty e^{-b\sqrt{u}}\,du = e^{-b}\left(1+\frac{2}{b}+\frac{2}{b^2}\right).$$

This proves the bound. The last assertion follows from the exponential power series. Notice also that $E(b)$ decreases with $b > 0$. □

Lemma 2.3. *Let $t = (A/(2n))^{2/3}$, $x = t^{-1/2} \geq 100$, and $X = nt = Ax/2$. For $z = t + iy$ with $|y| \leq t$ one has*

$$b(z) > 5x, \qquad |R(z)| \leq 8e^{-5x}, \qquad |e^{R(z)} - 1| \leq 16e^{-5x}. \tag{14}$$

Let $F_0(z) = \sqrt{z}\, e^{A/\sqrt{z}}/(2\pi)$ and $M(n) = t^{3/2}e^{3X}/(2\pi^{3/2}\sqrt{3X})$. Replacing $\mathcal{F}(e^{-z})$ by $F_0(z)$ in the coefficient integral over $[t - it, t + it]$ incurs a normalized error bounded by

$$\begin{aligned} &\frac{1}{M(n)} \left| \frac{1}{2\pi i} \int_{t-it}^{t+it} (\mathcal{F}(e^{-z}) - F_0(z)) e^{nz}\, dz \right| \\ &\qquad \leq \frac{96}{5} \sqrt{\frac{3X}{\pi}} e^{-5x} < 22\sqrt{x}\, e^{-5x} < \frac{1}{1000} X^{-5}. \end{aligned} \tag{15}$$

On the positive real axis, $e^{R(t)} \leq 101/100$.

Proof. Here $|\arg z| \leq \pi/4$ and $|z| \leq \sqrt{2}t$, whence

$$b(z) \geq 2\pi^{3/2} 2^{-1/4} \sqrt{1 - 1/\sqrt{2}}\, x > 5x.$$

The last numerical inequality follows entirely from rational comparisons:

$$\pi > \frac{157}{50}, \quad \sqrt{\pi} > \frac{177}{100}, \quad 2^{-1/4} > \frac{21}{25}, \quad \sqrt{1 - 1/\sqrt{2}} > \frac{27}{50},$$

and twice the product of these four lower bounds exceeds 5. For the last square-root bound, use $\sqrt{2} > 707/500$ and $1 - 500/707 > (27/50)^2$. The bounds on π and A used here and below are those in (7).

Since $b > 5x \geq 500$, formula (13) gives $E(b) \leq 8e^{-5x}$. For example, $b \geq 4$ and $e^{-b} < 1/2$ already bound its prefactor by 8. Also $e^{500} > 8000$, so $E(b) < 1/1000$. For $0 \leq v < 1$, the exponential series gives $e^v \leq (1 - v)^{-1}$. Therefore

$$|e^{R(z)} - 1| \leq e^{E(b)} - 1 \leq \frac{E(b)}{1 - E(b)} \leq 2E(b) \leq 16e^{-5x}.$$

This proves (14); on the real axis it also gives $e^{R(t)} \leq (1 - 1/1000)^{-1} < 101/100$.

Along the segment, $|z|^{1/2} \leq 2^{1/4}\sqrt{t} < (6/5)\sqrt{t}$, and $\mathrm{Re}(z^{-1/2}) \leq t^{-1/2}$. Hence

$$|F_0(z)e^{nz}| \leq \frac{6\sqrt{t}}{10\pi} e^{3X}.$$

Multiply this bound by the segment length $2t$, the inverse-coefficient factor $1/(2\pi)$, and $16e^{-5x}$, and divide by $M(n)$. The result is the first bound in (15). Because $\pi > 3$ and $X < (29/25)x$, the next inequality follows by squaring: $(96/5)^2(29/25) = 267264/625 < 22^2$.

Finally $x^{11/2}e^{-5x}$ decreases for $x \geq 100$. The elementary positive-term inequality $e^{500} > 500^8/8!$ yields

$$22 \left(\frac{29}{25}\right)^5 100^{11/2} \frac{8!}{500^8} = \frac{113713810056}{2384185791015625} < \frac{1}{1000}.$$

Together with $X < (29/25)x$, this proves the final comparison for the entire unbounded domain $x \geq 100$. □

3 An explicit approximation on the principal arc

Put

$$a = (A/2)^{2/3}, \qquad t = (A/(2n))^{2/3}, \qquad X = nt = an^{1/3}, \qquad F_0(z) = \frac{\sqrt{z}}{2\pi}\exp(A/\sqrt{z}), \tag{16}$$

and define

$$M(n) = \frac{t^{3/2}e^{3X}}{2\pi^{3/2}\sqrt{3X}} = \frac{ae^{3an^{1/3}}}{2\sqrt{3}\,\pi^{3/2}n^{7/6}}. \tag{17}$$

Principal powers are used throughout the slit plane $\mathbb{C} \setminus (-\infty, 0]$. The principal-arc integral for this model is

$$V(n) = \frac{1}{2\pi i}\int_{t-it}^{t+it} F_0(z)e^{nz}\,dz.$$

The purpose of this section is an explicit approximation to $V(n)$; the replacement of F_0 by the actual generating function uses the transformation estimate separately.

3.1 An exact steepest contour

Let $H(n) = (2\pi i)^{-1}\int_{\mathcal{C}} F_0(z)e^{nz}\,dz$, where $\mathcal{C}$ is a Hankel contour from negative infinity below the cut, passing to the right of zero, and returning to negative infinity above the cut. Its distance from zero is fixed and positive. Exponential decay at infinity and Cauchy's theorem justify deformation between such contours. No limiting collapse onto the essential singularity at zero is used.

Set $z = tw^2$. A suitable upper half of the contour is parametrized by

$$w = u + iv, \qquad v^2 = u^{-1} - u^2, \qquad y^2 = 3 - 4u^2 + u^{-1}, \qquad 0 < u \le 1, \quad 0 \le y < \infty. \tag{18}$$

The last expression decreases strictly with u, from infinity to zero, so it defines a unique $u = u(y)$. Direct substitution gives

$$w^2 + 2/w = 3 - y^2.$$

As $u \downarrow 0$, $\mathrm{Re}(tw^2) \to -\infty$ and $\mathrm{Im}(tw^2) \to 0^+$. The lower reflected path therefore completes an admissible Hankel contour. Since $\sqrt{z}\,dz = 2t^{3/2}w^2\,dw$, combining its two halves yields

$$H(n) = \frac{t^{3/2}e^{3X}}{\pi^2}\int_0^\infty e^{-Xy^2}g(y)\,dy, \qquad g(y) = \mathrm{Im}\big(w(y)^2w'(y)\big). \tag{19}$$

Both the inverse-Laplace factor and the factor $1/(2\pi)$ in F_0 are included in this normalization.

Lemma 3.1. *The real function g in (19) has an even holomorphic extension to $|y| < 1$. Writing $g(y) = \sum_{r\ge 0} g_{2r}y^{2r}$, one has*

$$g_0 = 1/\sqrt{3}, \qquad |g_{2r}| \le 43.$$

For real $y \ge 0$,

$$|g(y)| \le 1 + 4y^2 + y^4. \tag{20}$$

In particular, for $0 \le y \le 4/5$,

$$\left| g(y) - \sum_{r=0}^{10} g_{2r}y^{2r} \right| \le 120y^{22}. \tag{21}$$

Proof. The selected branch $w(0) = 1$, $w'(0) = i/\sqrt{3}$ satisfies

$$(w-1)^2(w+2) + y^2 w = 0. \tag{22}$$

On $|w-1| = 1$ the first summand has modulus at least 2, while the second has modulus at most $2|y|^2$. Rouché's theorem places precisely two roots inside this circle for $|y| < 1$. At zero these roots split analytically: substitute $w = 1 + yv$ and apply the implicit function theorem at $v = \pm i/\sqrt{3}$. Other possible multiple-root parameters satisfy $(3-y^2)^3 = 27$ and have modulus $\sqrt{3\sqrt{3}} > 1$. Consequently the selected branch continues holomorphically through the unit disk, with $|w-1| < 1$. There is no monodromy around zero because both local branches are already analytic there.

Differentiation of (22), with removable values interpreted by continuity, gives

$$w' = -\frac{yw^2}{w^3-1}, \qquad |w^2 w'| = \frac{|w|^{7/2}|w+2|^{1/2}}{|w^2+w+1|}.$$

The two roots of $w^2 + w + 1$ lie at distance $\sqrt{3}$ from 1. It follows that

$$|w^2 w'| < \frac{2^{9/2}}{(\sqrt{3}-1)^2} < 43.$$

For example, the last numerical inequality follows from $\sqrt{3} < 26/15$ and $\sqrt{512} < 68/3$, which give the sharper upper bound $85/2$.

To make the holomorphic extension explicit, write $h(y) = w(y)^2 w'(y)$ and define

$$g(y) = \frac{h(y) + h(-y)}{2i}.$$

For real y one has $w(-y) = \overline{w(y)}$ and $w'(-y) = -\overline{w'(y)}$, so this extension agrees with $\operatorname{Im} h(y)$. It is even and bounded in modulus by 43. Cauchy's estimates, first on radius $r < 1$ and then with $r \uparrow 1$, give $|g_{2r}| \le 43$. Summing the geometric remainder gives

$$\left| g(y) - \sum_{r=0}^{10} g_{2r} y^{2r} \right| \le \frac{43 y^{22}}{1-y^2} \le 120 y^{22}, \qquad 0 \le y \le 4/5.$$

For the global real bound, (18) implies

$$\frac{v}{y} = \frac{\sqrt{1+u+u^2}}{1+2u} =: b(u), \qquad |u'| = \frac{2yu^2}{1+8u^3} \le 2y.$$

Here $0 < b \le 1$, and logarithmic differentiation gives $|b'| = 3b/[2(1+u+u^2)(1+2u)] \le 3/2$. Thus $v \le y$ and $0 < v' \le 1 + 3y^2$. Finally

$$\operatorname{Im}(w^3) = v(4u^2 - u^{-1}) = v(3-y^2), \qquad g = \tfrac{1}{3}\frac{d}{dy}\{v(3-y^2)\}.$$

Taking absolute values proves (20). □

3.2 A rational polynomial and its error

For $0 \le r \le 10$, define the rational numbers

$$c_r = \frac{(-1)^r (1/2)_r}{3^r} [z^{2r}](1+z)^{r+5/2}(1+z/3)^{-r-1/2}, \qquad P(X) = \sum_{r=0}^{10} c_r X^{-r}, \tag{23}$$

where $(b)_r = b(b+1)\cdots(b+r-1)$ and $(b)_0 = 1$. For orientation,

$$c_0 = 1, \quad c_1 = -\frac{17}{36}, \quad c_2 = -\frac{35}{2592}, \quad c_3 = -\frac{1925}{279936}.$$

All eleven coefficients can be reproduced by exact rational arithmetic from (23).

Here is a derivation of this formula, including its normalization. Put $\delta = w - 1$. Equation (22) becomes $\delta = iy\sqrt{(1+\delta)/(3+\delta)}$ on the selected branch. Lagrange inversion and $h = (w^3)'/3$ give

$$[y^{2r}]h(y) = i(-1)^r 3^{-r-1/2} [z^{2r}](1+z)^{r+5/2}(1+z/3)^{-r-1/2}.$$

Since $g_{2r} = [y^{2r}]h/i$ and $\int_0^\infty e^{-Xy^2} y^{2r}\, dy = \frac{1}{2}\Gamma(r+1/2)X^{-r-1/2}$, this proves

$$c_r = \sqrt{3}\, g_{2r} \frac{\Gamma(r+1/2)}{\sqrt{\pi}} = \sqrt{3}\, g_{2r} (1/2)_r. \tag{24}$$

Lemma 3.2. *For every $X \ge 115$,*

$$\left| \frac{H(n)}{M(n)} - P(X) \right| \le BX^{-11} + 180 X^{-1/2} e^{-16X/25}, \qquad B = \frac{1443677610375}{1024} < 1410000000. \tag{25}$$

Proof. Integrating (21) on $[0, 4/5]$, enlarging this bound to the positive half-line, and normalizing by $\sqrt{\pi}/(2\sqrt{3X})$ bounds the local error by

$$120\sqrt{3}\,(1/2)_{11} X^{-11} < 210 (1/2)_{11} X^{-11} = BX^{-11}.$$

It remains to bound both Gaussian tails rather than assume g is positive; in fact g is eventually negative.

Set $d = 4/5$ and $J_r = \int_d^\infty y^{2r} e^{-Xy^2}\, dy$. Integration by parts gives

$$J_0 \le \frac{e^{-Xd^2}}{2Xd}, \qquad J_r = \frac{d^{2r-1} e^{-Xd^2}}{2X} + \frac{2r-1}{2X} J_{r-1} \quad (r \ge 1).$$

For $r \le 10$ each successive term relative to its predecessor is at most $19/(2Xd^2) < 13/100$. Therefore

$$J_r \le \frac{e^{-Xd^2} d^{2r-1}}{2X(1-13/100)}, \quad \sum_{r=0}^{10} J_r < \frac{2e^{-16X/25}}{X}, \quad J_0 + 4J_1 + J_2 < \frac{3e^{-16X/25}}{X}.$$

The coefficient bound $|g_{2r}| \le 43$ and (20) show that the combined tail error is at most $89 e^{-16X/25}/X$. Normalization multiplies it by $2\sqrt{3X/\pi} < 2\sqrt{X}$, so $180 X^{-1/2} e^{-16X/25}$ is an upper bound. □

3.3 Matching the vertical segment

Lemma 3.3. *For $X \geq 115$,*

$$\frac{|H(n) - V(n)|}{M(n)} \leq 5X^{-1/2}e^{-11X/25}. \tag{26}$$

Proof. Deform the Hankel contour to the vertical segment $t - it$ to $t + it$ together with the horizontal rays $z = t(x \pm i)$, $-\infty < x \leq 1$. The connections at negative infinity vanish by exponential decay. For $x \leq 0$,

$$\mathrm{Re}\{x + i + 2(x+i)^{-1/2}\} \leq x + \sqrt{2}, \qquad |x+i|^{1/2} \leq 1 + |x|.$$

Indeed the real part of $(x+i)^{-1/2}$ increases with $x \leq 0$ and is $1/\sqrt{2}$ at zero. For $0 \leq x \leq 1$, the real phase has derivative $1 - \mathrm{Re}(x+i)^{-3/2} \geq 1/2$. When $x \leq 1/\sqrt{3}$ its derivative is at least 1; on the remaining interval the subtracted real part is at most $\cos(3\pi/8) < 1/2$. The phase at 1 equals

$$1 + \sqrt{1 + \sqrt{2}} < 64/25.$$

The last inequality follows from $\sqrt{2} < 10/7$ and $17/7 < (39/25)^2$. Also $|x+i|^{1/2} \leq 2^{1/4} < 6/5$ there. The two pieces of either horizontal ray consequently have absolute integrals bounded by

$$(X^{-1} + X^{-2})e^{\sqrt{2}X} \quad \text{and} \quad \frac{12}{5X}e^{64X/25},$$

respectively, after removal of the common dimensional prefactor. Inserting both rays, the inverse-Laplace factor, and $1/(2\pi)$ from F_0 yields

$$\frac{|H - V|}{M} \leq \sqrt{\frac{3}{\pi X}} \left\{ \frac{12}{5}e^{-11X/25} + (1 + X^{-1})e^{-(3-\sqrt{2})X} \right\} \leq 5X^{-1/2}e^{-11X/25}.$$

□

Proposition 3.4. *The explicitly defined polynomial (23) satisfies*

$$\left| \frac{V(n)}{M(n)} - P(X) \right| < \frac{1}{500}X^{-5} \qquad (X \geq 115). \tag{27}$$

Proof. Combine Lemmas 3.2 and 3.3. The rational inequality $1000B < 115^6$ gives $BX^{-11} < X^{-5}/1000$. Each exponential term is less than $X^{-5}/10000$. For a completely rational check of the latter assertions, the positive-term Taylor sum for e^q, through degree 160, proves

$$e^{16 \cdot 115/25} > 10000 \cdot 180 \cdot 115^5, \qquad e^{11 \cdot 115/25} > 10000 \cdot 5 \cdot 115^5.$$

Since $X^{9/2} \leq X^5$ and $X^5 e^{-cX}$ decreases for $X \geq 115$ for both values of c, these endpoint checks prove the claims on the full interval. The exact arithmetic verifier described in the reproducibility appendix checks these displayed inequalities and (23). Their sum is smaller than $X^{-5}/500$. □

4 Curvature of the explicit approximation

Define the positive smooth comparison function

$$Q(n) = M(n)P(an^{1/3}).$$

The following estimates concern this explicit function only. No unknown coefficient error is differentiated.

Proposition 4.1. *For $X = an^{1/3} \geq 115$,*

$$P(X) > 33/50, \qquad X|P'(X)| < 35/100, \qquad X^2|P''(X)| < 72/100, \tag{28}$$

and

$$-(\log Q(n))'' \geq \frac{3}{5} a n^{-5/3}. \tag{29}$$

If $a(n-1)^{1/3} \geq 115$, then

$$2\log Q(n) - \log Q(n-1) - \log Q(n+1) \geq \frac{3}{5} a (n+1)^{-5/3}. \tag{30}$$

Proof. From (24) and Lemma 3.1, $|c_r| \leq 76(1/2)_r$. Define the rational sums

$$S_0 = \sum_{r=1}^{10} \frac{76(1/2)_r}{115^r}, \quad S_1 = \sum_{r=1}^{10} \frac{76r(1/2)_r}{115^r}, \quad S_2 = \sum_{r=1}^{10} \frac{76r(r+1)(1/2)_r}{115^r}.$$

Exact summation gives $S_0 < 34/100$, $S_1 < 35/100$, and $S_2 < 72/100$. These rational inequalities can also be checked directly by putting the ten summands over their common denominator. All the bounds decrease with X, proving (28). Writing $L(X) = \log P(X)$ and using $X' = X/(3n)$ gives

$$n^2 \frac{d^2}{dn^2} L(X(n)) = \frac{X^2 L''(X) - 2X L'(X)}{9}.$$

Its absolute value is at most

$$\frac{1}{9}\left\{ \frac{S_2}{1-S_0} + \frac{S_1^2}{(1-S_0)^2} + \frac{2S_1}{1-S_0} \right\} < \frac{1}{2}.$$

The final strict inequality already follows from the displayed rational upper bounds for S_0, S_1, and S_2. The verifier also checks it with their exact rational values. Since

$$(\log M(n))'' = -\frac{2X}{3n^2} + \frac{7}{6n^2},$$

we obtain

$$-(\log Q(n))'' \geq \frac{(2/3)X - 5/3}{n^2} \geq \frac{3}{5}\frac{X}{n^2} = \frac{3}{5} a n^{-5/3}.$$

The second inequality holds for $X \geq 25$ and thus on the stated domain. Finally, for twice continuously differentiable f,

$$2f(n) - f(n-1) - f(n+1) = -\int_{-1}^{1} (1-|u|) f''(n+u)\, du.$$

Apply this identity to $f = \log Q$, use the decreasing lower bound $(3/5)a(n+u)^{-5/3} \geq (3/5)a(n+1)^{-5/3}$, and note that the triangular kernel integrates to one. □

5 Uniform bounds away from the saddle

Define

$$\rho(t,\theta) = \frac{|\mathcal{F}(\mathrm{e}^{-t+\mathrm{i}\theta})|}{\mathcal{F}(\mathrm{e}^{-t})}.$$

Each factor gives the exact identity

$$\rho(t,\theta) = \prod_{j\ge 1} \left(1 + \frac{\sin^2(j^2\theta/2)}{\sinh^2(j^2 t/2)}\right)^{-1/2}. \tag{31}$$

Every factor is at most one, so any subset gives an upper bound.

Lemma 5.1. *For $0 < t \le 10^{-4}$ and $x = t^{-1/2}$,*

$$\int_t^\pi \rho(t,\theta)\,d\theta \le \frac{128t}{x}\mathrm{e}^{-38x/125} + \frac{1}{100}\left(\mathrm{e}^{-3x/5} + (110t)^9\right) + \pi\,10^{-25}(10000t)^8. \tag{32}$$

Proof. First suppose $t \le \theta \le 1/100$. For $j^2\theta \le 1$, power-series bounds on $0 \le u \le 1/2$ give $\sin u/u \ge 23/24$ and $\sinh u/u \le 261/250$. For example, the nonconstant tail of $\sinh u/u$ is at most $(1/24)/(1-1/80) = 10/237 < 11/250$. The squared ratio of these two bounds exceeds $c = 21/25$, whence

$$\rho(t,\theta) \le [1 + c(\theta/t)^2]^{-\lfloor \theta^{-1/2}\rfloor/2}. \tag{33}$$

Put $r = \theta/t$, $h(r) = r^{-1/2}\log(1+cr^2)$, and $v = cr^2$. Then

$$h'(r) = \frac{4v/(1+v) - \log(1+v)}{2r^{3/2}}.$$

The numerator increases up to $v = 3$ and decreases thereafter. At $v = 21/25$ and $v = 336/25$ it exceeds 1. Consequently, for $1 \le r \le 4$, $h'(r) \ge 1/16$, $h(1)/2 > 38/125$, and $\log(1+cr^2)/2 < 1.34$, with $\mathrm{e}^{1.34} < 4$. Using $\lfloor\theta^{-1/2}\rfloor \ge \theta^{-1/2} - 1$ in (33) and integrating an exponential proves

$$\int_t^{4t} \rho(t,\theta)\,d\theta \le 4t\mathrm{e}^{-38x/125}\int_1^\infty \mathrm{e}^{-x(r-1)/32}\,dr = \frac{128t}{x}\mathrm{e}^{-38x/125}.$$

For $4t \le \theta \le 1/100$, $\lfloor\theta^{-1/2}\rfloor \ge (9/10)\theta^{-1/2}$. The numerator in h' is positive initially, increases until $v = 3$, then decreases to negative infinity. Thus h has a single maximum, and its minimum on a closed positive interval is at an endpoint. At $r = 4$, $(9/20)xh(4) > (3/5)x$. At $\theta = 1/100$ it equals $(9/2)\log(1 + c/(10000t^2))$. Since $100/\sqrt{c} < 110$, this gives

$$\int_{4t}^{1/100} \rho(t,\theta)\,d\theta \le \frac{1}{100}\left(\mathrm{e}^{-3x/5} + (110t)^9\right).$$

It remains to prove a uniform bound on $1/100 \le \theta \le \pi$. Set $\alpha = \theta/(2\pi)$. Since $2\pi < 629/100$, $\alpha > 1/629 > 158/100000$. Cover the larger interval $[158/100000, 1/2]$ by the 49842 closed intervals $I_k = [k/B, (k+1)/B]$, where $B = 100000$ and $158 \le k < 50000$. For $1 \le j \le 40$, compute exactly

$$d_{j,k} = \min_{\alpha\in I_k} \|j^2\alpha\|, \qquad a_{j,k} = \left(\frac{20000 d_{j,k}}{j^2}\right)^2.$$

Here $\|u\|$ is distance to the nearest integer. The minimum is zero if the image interval contains an integer, and otherwise is the smaller endpoint distance. Thus every $d_{j,k}$ and $a_{j,k}$ is rational. The following two integer checks hold for every k:

$$\prod_{j=1}^{40}(1 + \lfloor a_{j,k}\rfloor) \ge 10^{50}, \qquad \#\{j : a_{j,k} \ge 1\} \ge 16. \tag{34}$$

The complete verification algorithm appears in Appendix A. Its minimum product is

$$114916374286660554473365891018961431890522931200000000,$$

and the minimum count is 20, giving margins in both comparisons.

To explain why this finite calculation proves a universal statement, let $t_0 = 10^{-4}$. The inequalities $|\sin(\pi u)| \geq 2\|u\|$ and $\sinh(j^2 t_0/2) \leq j^2 t_0$ for $j \leq 40$ show that $a_{j,k}$ is a lower bound on the corresponding squared ratio in (31). The latter sinh bound follows from $j^2 t_0/2 \leq 0.08$ and $\sinh u \leq 2u$ on $[0, 1/2]$, already implied by the power-series bound. Convexity gives $\sinh(j^2 t/2) \leq (t/t_0)\sinh(j^2 t_0/2)$. For $U = (t_0/t)^2 \geq 1$ and $a \geq 0$,

$$\log(1 + Ua) - \log(1 + a) \geq \frac{a}{1 + a} \log U,$$

as follows by integration in $\log U$. Equation (34) and $\sum_j a_{j,k}/(1 + a_{j,k}) \geq 8$ now give

$$\rho(t, \theta) \leq 10^{-25}(10000t)^8 \quad (0 < t \leq 10^{-4},\ 1/100 \leq \theta \leq \pi). \tag{35}$$

Integration over an interval of length at most π finishes the proof. All elementary numerical comparisons above are rationally verified as described in Appendix B. □

Proposition 5.2. *The contribution to $s(n)$ of the two arcs $t \leq |\theta| \leq \pi$, divided by $M(n)$, has absolute value less than $(1/40)X^{-5}$ whenever $t \leq 10^{-4}$ and $x \geq 100$.*

Proof. Cauchy's coefficient formula and symmetry bound the contribution by

$$\frac{\mathrm{e}^{nt}\mathcal{F}(\mathrm{e}^{-t})}{\pi} \int_t^\pi \rho(t, \theta)\, d\theta.$$

The exact transformation and the definition of M show that the normalizing factor is

$$\frac{\mathrm{e}^{nt}\mathcal{F}(\mathrm{e}^{-t})}{\pi M(n)} = \mathrm{e}^{R(t)} \frac{\sqrt{3X/\pi}}{t}.$$

Use $\mathrm{e}^{R(t)} \leq 101/100$, $\sqrt{3X/\pi} < (11/10)\sqrt{x}$, and $X \leq (29/25)x$. After multiplying (32) by this factor and by X^5, the four terms are bounded by

$$C\left(128x^{9/2}\mathrm{e}^{-38x/125} + \frac{x^{15/2}}{100}\mathrm{e}^{-3x/5} + \frac{110^9}{100}x^{-21/2} + \frac{22}{7}10^7 x^{-17/2}\right),$$

$$C = \frac{1111}{1000}\left(\frac{29}{25}\right)^5.$$

Every term decreases for $x \geq 100$: for a term $x^p\mathrm{e}^{-cx}$, its logarithmic derivative is $p/x - c < 0$ on this domain, and the other powers have negative exponents. At $x = 100$, rational bounds for the exponentials give a total less than $0.0188 < 1/40$. The verifier performs this comparison with exact fractions. □

6 Combining the estimates and controlling second differences

Proposition 6.1. *For every integer $m \geq N_0$, with $X = am^{1/3}$,*

$$\left|\frac{s(m)}{M(m)} - P(X)\right| < \frac{1}{10}X^{-5}. \tag{36}$$

Proof. Cauchy's coefficient formula on the circle $|q| = e^{-t}$ gives

$$s(m) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \mathcal{F}(e^{-(t+i\theta)}) e^{m(t+i\theta)}\, d\theta.$$

Split the integral at $|\theta| = t$. Lemma 2.3 bounds the error in replacing $\mathcal{F}(e^{-z})$ by $F_0(z)$ on the central segment by $M(m)X^{-5}/1000$. Proposition 3.4 bounds the difference between this model integral and $M(m)P(X)$ by $M(m)X^{-5}/500$. Proposition 5.2 bounds the two remaining arcs together by $M(m)X^{-5}/40$. All their hypotheses hold by (8), and

$$\frac{1}{1000} + \frac{1}{500} + \frac{1}{40} = \frac{7}{250} < \frac{1}{10}.$$

This proves the assertion. □

Proposition 6.2. *For every integer $n \geq N_0 + 1$, $D(n) > 0$.*

Proof. Let $Q(m) = M(m)P(am^{1/3})$. Since $P > 33/50$ by Proposition 4.1, (36) gives

$$\left| \frac{s(m)}{Q(m)} - 1 \right| < \frac{5}{33} X(m)^{-5} \qquad (m \geq N_0).$$

This bound is less than $1/11$. The elementary inequality $|\log(1+u)| \leq |u|/(1-|u|)$ for real $|u| < 1$ therefore gives

$$|\log s(m) - \log Q(m)| < \frac{1}{6} X(m)^{-5}, \qquad m \geq N_0. \tag{37}$$

Here the factor $11/10$ bounds $(1-|u|)^{-1}$, and $(11/10)(5/33) = 1/6$.

For $n \geq N_0 + 1$, all three indices $n-1, n, n+1$ satisfy (37). Since $X(m)^{-5} = a^{-5} m^{-5/3}$ decreases, the error in their logarithmic second difference is less than

$$\frac{2}{3} a^{-5} (n-1)^{-5/3}.$$

On the other hand, (30) gives

$$2\log Q(n) - \log Q(n-1) - \log Q(n+1) \geq \frac{3}{5} a (n+1)^{-5/3}.$$

This lower bound strictly exceeds the preceding error. To see this using simple rational inequalities, (7) implies

$$a^6 = (A/2)^4 > (23/20)^4 > 17/10.$$

For $n \geq 100$, Bernoulli's inequality with exponent $5/3$ gives

$$\left(\frac{n-1}{n+1} \right)^{5/3} \geq 1 - \frac{10}{3(n+1)} > \frac{9}{10}.$$

Consequently

$$\frac{3}{5} a^6 \left(\frac{n-1}{n+1} \right)^{5/3} > \frac{3}{5} \frac{17}{10} \frac{9}{10} = \frac{459}{500} > \frac{2}{3}.$$

It follows that $2\log s(n) - \log s(n-1) - \log s(n+1) > 0$. All square-partition numbers are positive, because the all-ones partition always exists, so exponentiating proves $D(n) > 0$. Notice that the unknown error in (36) was never differentiated: its three absolute values were bounded directly. □

7 Exact finite verification and the sharp boundary

The remaining computation covers every $D(n)$ for $1 \le n \le N_0 = 1{,}200{,}000$. Because $D(N_0)$ uses $s(N_0+1)$, the coefficient table is computed through $B = N_0 + 1$. All coefficients and all differences use arbitrary-precision integers.

7.1 The finite product and its correctness invariant

Parts greater than B cannot affect a coefficient of degree at most B. Thus it is enough to compute

$$\prod_{1 \le j \le \lfloor \sqrt{B} \rfloor} (1 - q^{j^2})^{-1} \pmod{q^{B+1}}.$$

Let $P_r(k)$ count partitions of k using only $1, 4, \ldots, r^2$. With $P_r(k) = 0$ for $k < 0$, the exact recurrence is

$$P_r(k) = P_{r-1}(k) + P_r(k - r^2). \tag{38}$$

The first summand counts partitions with no r^2 part. Removing one r^2 from a partition having at least one such part gives precisely the second summand. Starting from $P_0(0) = 1$ and $P_0(k) = 0$ for $k > 0$, the following ascending updates implement this recurrence:

```
a = [0] * (B + 1)
a[0] = 1
for j in range(1, isqrt(B) + 1):
    square = j*j
    for k in range(square, B + 1):
        a[k] += a[k-square]
```

After processing $j = r$, the invariant $a[k] = P_r(k)$ follows by induction from (38). Ascending k uses the already updated $P_r(k - r^2)$, allowing repeated parts. No ordering of the parts is introduced. This proves that the completed table equals $s(k)$ for every $0 \le k \le B$.

7.2 An independent coefficient recurrence

Formal logarithmic differentiation of (1) yields

$$\frac{q\mathcal{F}'(q)}{\mathcal{F}(q)} = \sum_{j \ge 1} \frac{j^2 q^{j^2}}{1 - q^{j^2}} = \sum_{j \ge 1} \sum_{\ell \ge 1} j^2 q^{\ell j^2}$$
$$= \sum_{m \ge 1} b(m) q^m, \qquad b(m) = \sum_{j^2 \mid m} j^2.$$

Each formal coefficient involves only finitely many terms, so these operations need no analytic convergence argument. Multiplying by $\mathcal{F}$ and comparing coefficients gives

$$n s(n) = \sum_{m=1}^{n} b(m) s(n-m), \qquad s(0) = 1. \tag{39}$$

This determines the coefficients uniquely by induction. An independent implementation agreed with the product computation at every index $0 \le n \le 10001$, including all exceptional indices and the threshold. Every division in (39) was checked for zero remainder. Its quadratic cost is the reason this distinct-algorithm check was confined to 10001; the entire interval through 1,200,001 was recomputed by a separate implementation of the product algorithm.

7.3 Results and exact boundary values

The complete finite computation gives exactly 421 negative differences, exactly six zero differences, and 1,199,573 positive differences on $1 \le n \le N_0$. The negative indices are those of Table 1; the zero indices are $1, 2, 5, 6, 10, 14$. In particular every integer $1042 \le n \le N_0$ has $D(n) > 0$. The four boundary coefficients, obtained independently by both recurrences, are

n	$s(n)$
1040	5,906,030,608
1041	5,963,155,167
1042	6,020,833,213
1043	6,079,043,298

They give

$$\begin{aligned} D(1041) &= 5{,}963{,}155{,}167^2 - 5{,}906{,}030{,}608 \cdot 6{,}020{,}833{,}213 \\ &= -5{,}695{,}922{,}185{,}615, \\ D(1042) &= 6{,}020{,}833{,}213^2 - 5{,}963{,}155{,}167 \cdot 6{,}079{,}043{,}298 \\ &= 154{,}125{,}878{,}482{,}603. \end{aligned}$$

The accompanying certificate records every negative and zero index, the interval limits, the exact boundary integers, and cryptographic digests identifying the coefficient and sign tables. The independent auditor reconstructs the coefficient table from scratch, checks every asserted sign and count, expands the compressed negative set, and verifies the digests. A digest identifies data; it is not a substitute for this recomputation or for the correctness invariant above. The numeric checkpoint supports resumption after interruption, while the independent auditor reads no checkpoint. The certificate appendix specifies the encoding and the reproduction commands.

Proof of Theorem 1.1. The exact finite computation covers all integers $1 \le n \le N_0$, and Proposition 6.2 covers every integer $n \ge N_0 + 1$. Thus there is no missing index between the finite and infinite ranges. Together they prove the stated complete negative and equality sets and strict positivity everywhere else. In particular, $D(n) > 0$ for all $n \ge 1042$, whereas $D(1041) < 0$. The definition of T in (2) therefore gives $T = 1042$. □

8 Verification scope

The proof has two different computational roles. The finite partition calculation evaluates exact integers after a proved recurrence. The analytic certificates check rational inequalities and finite interval coverings that occur inside explicit bounds for the entire infinite tail. Numerical checks of the transformation at sample complex arguments are useful normalization tests, but they are not used as proof of the identity. The identity is derived analytically in Section 2.

The analytic estimates, implementations and manuscript were developed by GPT-6 Astra, an AI system, which also executed the verification. Separate internal reviews examined the transformation, contour estimates, finite calculations, and their integration. These checks are internal adversarial review, not external mathematical peer review. No Lean or other proof-assistant theorem was compiled, and no formal-verification claim is made. The proof and its accompanying verifiers are provided for independent inspection and reproduction.

A Exact arithmetic certificates and their verifiers

All arithmetic needed for the proof can be performed with Python 3.9 or later using only its standard library. Integers have arbitrary precision, and rational comparisons use `fractions.Fraction`. Floating-point output, where present, is descriptive only and is never used to decide an inequality. The optional high-precision transformation diagnostic is separate and is not part of the proof.

A.1 The far-arc certificate in full

The following complete algorithm checks the finite predicates used for the far arcs. It contains no trigonometric evaluations. For a grid cell $[k/B, (k+1)/B]$, the image under multiplication by j^2 has endpoints L/B and U/B. If its endpoints have different integer parts, it contains an integer and its minimum distance to an integer is zero. Otherwise the minimum is the smaller endpoint distance. An endpoint equal to an integer also gives zero in the displayed formula.

```
B, J, target = 100000, 40, 10**50
minimum_product = None
minimum_strong = J
worst_cell = None
for k in range(B // 629, B // 2):
    product, strong = 1, 0
    for j in range(1, J + 1):
        square = j*j
        left, right = square*k, square*(k+1)
        if left // B != right // B:
            d = 0
        else:
            d = min(left % B, B - right % B)
        numerator = (20000*d)**2
        denominator = (B*square)**2
        product *= 1 + numerator // denominator
        strong += numerator >= denominator
    if product < target or strong < 16:
        raise ArithmeticError(k, product, strong)
    if minimum_product is None or product < minimum_product:
        minimum_product, worst_cell = product, k
    minimum_strong = min(minimum_strong, strong)
print(minimum_product, worst_cell, minimum_strong)
```

There are 49842 cells, including their endpoints, and the output is

$$11491637428666055447336589101896143189052293120000000,$$
$$24995, \qquad 20.$$

The actual proof uses only the weaker product lower bound 10^{50} and the count lower bound 16. The grid is a proof of inequalities on closed intervals, not a sampling of their midpoints. The analytic extension from t_0 to every $0 < t \le t_0$ is proved in the minor-arc section. The packaged version is `code/verify_minor_arcs.py`; its output is `certificates/minor_arcs.json`.

A.2 Finite coefficients and identification of the data

The basic finite computation is the following integer algorithm, with $L = 1200000$:

```
from math import isqrt
L = 1200000
s = [1] + [0]*(L+1)
```

```
for j in range(1, isqrt(L+1)+1):
    for n in range(j*j, L+2):
        s[n] += s[n-j*j]
negative, zero = [], []
for n in range(1, L+1):
    D = s[n]*s[n] - s[n-1]*s[n+1]
    if D < 0:
        negative.append(n)
    elif D == 0:
        zero.append(n)
```

The increasing order of the inner loop is essential: it permits repeated copies of the current square. Its partition-counting invariant and the independent logarithmic-derivative recurrence are proved in the finite verification section. The coefficient table ends at $s(L+1)$, so the last difference $D(L)$ has both neighbors available.

The producer `code/finite_verify.py` also computes the independent recurrence through $s(10001)$ and records every exceptional index. The separate auditor `code/verify_finite_certificate.py` imports no producer code and reads no checkpoint. It recomputes the whole table, tests all signs, checks the boundary values, and independently constructs the recurrence weights by testing square divisors. The independent mathematical algorithms are compared through index 10001; both full-range executions use dynamic programming.

The final coefficient table has SHA-256 digest

`2fe028244688646268939f4a327e6ceb386adc131ead185912fa8a30b572d35a`

Its canonical encoding is specified completely as follows. For each coefficient in index order from 0 through 1200001, write its minimal unsigned little-endian byte string, using one zero byte for the integer zero. Precede that string by its byte count encoded as a four-byte unsigned little-endian integer. Concatenate all records before hashing. The sign digest uses one ASCII character `-`, `0`, or `+` for each successive $D(n)$, starting at $n = 1$. These hashes identify the computed data; recomputation and the proved algorithm, rather than a hash alone, establish its mathematical meaning.

The binary checkpoint files use this same integer encoding together with a bound, the next square to process, and an integrity checksum. They support resumption after interruption. They are not trusted by the separate auditor. The certificate is `certificates/finite_1200000.json`, and the observed audit result is recorded in `certificates/audit_1200000.txt`.

A.3 Verifier map and reproduction

The essential arithmetic checks are divided into the following small programs. The mathematical arguments giving meaning to the checks are in the body of the paper and in Appendix B.

`code/verify_global_constants.py`	π, A, transformed error
`code/verify_major_arc.py`	c_r, Gaussian error, curvature
`code/verify_minor_arcs.py`	closed far-arc intervals
`code/verify_minor_constants.py`	minor-arc numerical constants
`code/finite_verify.py`	exact partition coefficients
`code/verify_finite_certificate.py`	separate finite audit

The first four programs run without arguments. A clean finite reproduction can be run in PowerShell as follows; the backticks continue the same command onto the next line.

```
python code/finite_verify.py `
  --limit 1200000 --independent-through 10001 `
  --output certificates/finite_1200000.json
```

```
python code/verify_finite_certificate.py ‘
  certificates/finite_1200000.json
```

The optional checkpoint and resumption arguments are documented in the package README. No proof-assistant compilation is claimed.

B Rational bounds for every numerical constant

B.1 Elementary interval constructions

For a positive rational q and a positive integer scale S, let $k = \lfloor S\sqrt{q} \rfloor$. Integer square-root arithmetic computes k without a floating-point square root, and

$$\frac{k}{S} \le \sqrt{q} < \frac{k+1}{S}.$$

The verifier uses $S = 10^{18}$. For $0 < x < 1$, the alternating series

$$\arctan x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{2k+1}$$

lies between a partial sum and that partial sum plus its next term. Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ follows by the tangent addition formula; the angles lie in the interval selecting $\pi/4$ before multiplication by 4. Forty terms at each argument prove

$$\frac{157}{50} < \pi < \frac{22}{7}.$$

Alternatively, the upper bound follows by integrating the positive function $x^4(1-x)^4/(1+x^2)$ on $[0,1]$, whose integral is $22/7 - \pi$. Eight terms of the integrated alternating geometric series for $(1+x^2)^{-1}$ already give $\pi > 3$.

For the zeta value, monotonicity of $x^{-3/2}$ gives, with $N = 1000$,

$$\sum_{k=1}^{N} k^{-3/2} + \frac{2}{\sqrt{N+1}} < \zeta(3/2) < \sum_{k=1}^{N} k^{-3/2} + \frac{2}{\sqrt{N}}.$$

Apply the rational square-root intervals term by term and then to $A = (\sqrt{\pi}/2)\zeta(3/2)$. Exact rational addition proves

$$\frac{231}{100} < A < \frac{58}{25}, \qquad (A/2)^4 > \frac{17}{10}.$$

These deliberately coarse enclosures are sufficient for all the threshold and error comparisons; no digits of an uncertified zeta evaluation enter the proof.

For $q > 1$, set $z = (q-1)/(q+1)$. The positive series for its logarithm and an elementary geometric bound give

$$2\sum_{k=0}^{N-1} \frac{z^{2k+1}}{2k+1} \le \log q \le 2\sum_{k=0}^{N-1} \frac{z^{2k+1}}{2k+1} + \frac{2z^{2N+1}}{(2N+1)(1-z^2)}. \tag{40}$$

The minor-constant verifier uses $N = 160$. For a positive rational a,

$$e^{-a} \le \left(\sum_{k=0}^{N} \frac{a^k}{k!}\right)^{-1}. \tag{41}$$

Every summand is rational and positive. Degrees 160, 180, or 240 are used by the respective verifiers; an upper estimate of the omitted tail is unnecessary for this direction of inequality.

B.2 The explicit minor-arc normalization

The following comparisons, all checked by (40) or rational arithmetic, are the constants appearing in the near and intermediate arcs:

$$\left(\frac{23/24}{261/250}\right)^2 > \frac{21}{25}, \quad \frac{1}{2}\log\frac{46}{25} > \frac{38}{125}, \quad \frac{1}{2}\log\frac{361}{25} < \frac{67}{50} < \log 4,$$
$$\frac{4(21/25)}{1+21/25} - \log\frac{46}{25} > 1, \quad \frac{4(336/25)}{1+336/25} - \log\frac{361}{25} > 1,$$
$$\frac{9}{40}\log\frac{361}{25} > \frac{3}{5}, \qquad \frac{10000}{21/25} < 110^2.$$

For the real transformation remainder, $x = t^{-1/2} \geq 100$ implies $b = 2\pi^{3/2}x > 1000$. The first three Taylor terms show $e^{1000} > 10000$. Substitution of $e^{-b} < 1/10000$ and $b > 1000$ in the explicit bound $E(b)$ gives $E(b) < 1/1000$. Since $e^u \leq (1-u)^{-1}$ for $0 \leq u < 1$, it follows that $e^{R(t)} < 101/100$. Also $A < 58/25$ and $\pi > 3$ give $\sqrt{3X/\pi} \leq (11/10)\sqrt{x}$.

After multiplication by X^5 and the substitution $X < 29x/25$, the normalized minor-arc bound is at most the sum of

$$128Cx^{9/2}e^{-38x/125}, \qquad \frac{C}{100}x^{15/2}e^{-3x/5},$$
$$\frac{C}{100}110^9x^{-21/2}, \qquad C\frac{22}{7}10^{-25}10000^8x^{-17/2}, \qquad C = \frac{1111}{1000}\left(\frac{29}{25}\right)^5.$$

For a term x^pe^{-cx}, the logarithmic derivative is $p/x - c$. Both exponential terms therefore decrease on $x \geq 100$, as do the two negative powers. Applying (41) at $x = 100$ gives the following rational upper bounds:

term	upper bound
near	4683836574539/250000000000000
intermediate exponential	41/200000000000000
intermediate power	11004445583/200000000000000
far	36669/50000000000000

Their unrounded rational sum is bounded above by

$$\frac{9395184629827}{500000000000000} = 0.018790369259654 < \frac{1}{40}.$$

The displayed individual upper bounds were rounded upward separately; the total was rounded upward from the exact rational sum, so adding the displayed rounded entries is not the definition of the displayed total. The monotonicity argument proves the bound for every real $x \geq 100$, not merely at the checked endpoint.

B.3 Major-arc arithmetic

The coefficients c_r are specified by the finite rational formula (23). Generalized binomial coefficients are computed as $\binom{b}{k} = \prod_{j=0}^{k-1}(b-j)/(j+1)$. This also gives exact values of the three ten-term sums S_0, S_1, S_2 used in the curvature bound. The Gaussian and horizontal exponential comparisons use (41); their endpoint exponents and their monotonicity domains are stated in the major-arc proof. The verifier does not differentiate or approximate an unknown remainder. Its certificate records every rational coefficient and sum, so the numerical portion can be repeated without a symbolic algebra system.

References

[1] A. Roy, *Log-Concavity and the Multiplicative Properties of Restricted Partition Functions*, Annals of Combinatorics, published online 29 September 2026. doi:10.1007/s00026-026-00847-5.

[2] A. Roy, *Log-Concavity and the Multiplicative Properties of Restricted Partition Functions*, arXiv:2404.03153v4, 10 May 2025. https://arxiv.org/abs/2404.03153v4. Earlier versions 1–3 list B. Benfield and A. Roy.

[3] C. O'Sullivan, *Detailed asymptotic expansions for partitions into powers*, International Journal of Number Theory **19**(9) (2023), 2163–2196. https://arxiv.org/abs/2205.13468v2.

[4] B. Benfield, M. Paul, and A. Roy, *Turan inequalities for k-th power partition functions*, Journal of Mathematical Analysis and Applications **529**(1) (2024), 127678. doi:10.1016/j.jmaa.2023.127678.

[5] D. Zagier, *Power partitions and a generalized eta transformation property*, Hardy–Ramanujan Journal **44** (2021), 1–18. https://hrj.episciences.org/8932/pdf.

[6] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A346160, inspected 23 October 2026. https://oeis.org/A346160.